Q.22 Discuss various used of Urea.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain classification of mass transfer operations with suitable examples.

Q.24 Explain classification of heat exchangers and evaporators used in chemical industry.

Q.25 Describe urea manufacturing process with flow sheet and list unit processes and unit operations involved.

No. of Printed Pages : 4

Roll No. ....

1st Year / Chemical Engineering
Subject : Introduction to Chemical Engineering

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Head transfer by electromagnetic waves is known as:

a) Conduction b) Convection

c) Radiation d) Diffusion

Q.2 Newton's law of cooling is mainly applicable to:

a) Conduction b) Convection

c) Radiation d) Diffusion

Q.3 SI unit of diffusivity is:

a) m^2/S b) Kg/m^3

c) W/m*K d) J/kg*K

Q.4 Reaction occurring at interface of two phases is:

a) Homogeneous b) Heterogeneous

c) Elementary d) Irreversible

Q.5 Molecularity of a reaction cannot be:

a) One b) Two

c) Three d) Fractional

Q.6 Reactor operating under steady state conditions is:

a) Bath reactor b) Autoclave

c) Plug flow reactor d) Bomb reactor

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define convection heat transfer.

Q.8 What is grey body?

Q.9 Write chemical formula of Urea.

Q.10 Define mass transfer.

Q.11 Define half life period.

Q.12 What is unit operation.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain Fourier's law of heat conduction.

Q.14 Explain Newton's law of cooling with example.

Q.15 Classify boilers and condensers used in chemical industry. (Names only)

Q.16 Define diffusion and explain factors affecting diffusion rate.

Q.17 Classify equipment used for distillation and drying.

Q.18 Differentiate between homogeneous and heterogeneous reactions.

Q.19 Explain reversible and irreversible reactions with examples.

Q.20 Calculate the value of rate constant for first order reaction if value of half life period is 30 sec.

Q.21 Explain classification of chemical reactors used in industry.